GENERAL NOTES: UNLESS OTHERWISE SPECIFIED

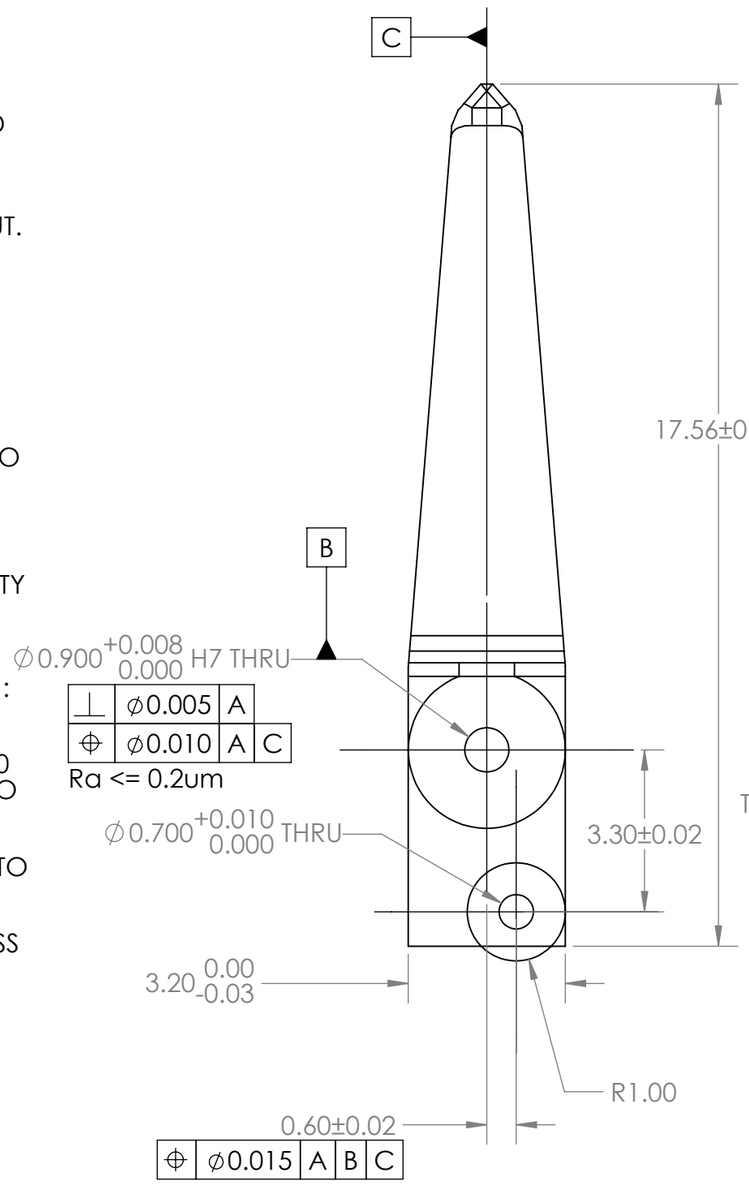
1. GENERAL & STANDARDS COMPLIANCE:
1.1 INTERPRET DRAWING IN ACCORDANCE WITH ASME Y14.5-2018 AND ISO 8015.
1.2 3D CAD DATASET (W3-SURG-GRASPER-5.1.STEP) GOVERNS NOMINAL GEOMETRY FOR UNTOLERANCED SURFACES.
1.3 GENERAL TOLERANCES: LINEAR +/- 0.02 mm, ANGULAR +/- 0.2 DEG (FINE PRECISION MACHINING).
1.4 THIRD ANGLE PROJECTION APPLIES THROUGHOUT. ALL UNITS IN MILLIMETERS.

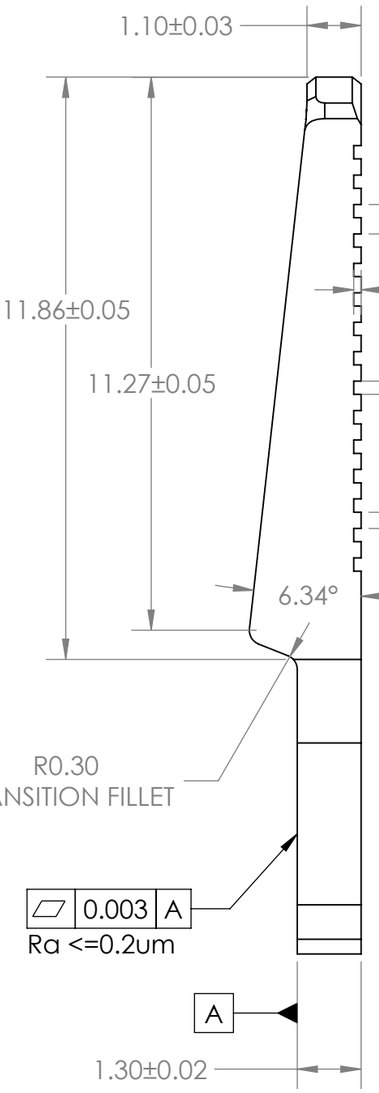
2. MATERIAL & BIOCOMPATIBILITY SPECIFICATIONS:
2.1 MATERIAL: MEDICAL GRADE TITANIUM ALLOY Ti-6Al-4V ELI PER ASTM F136 / ISO 5832-3
(ALTERNATIVE: VACUUM MELTED STAINLESS STEEL 316LVM PER ASTM F138 / ISO 5832-1).
2.2 FULL MATERIAL CERTIFICATION (MTR) AND LOT TRACEABILITY MANDATORY IN ACCORDANCE WITH ISO 13485.
2.3 100% HOMOGENEOUS SUBTRACTIVE CNC MACHINING ONLY. NO METAL ADDITIVE MANUFACTURING (SLM/EBM) OR POWDER-METALLURGY PERMITTED TO PREVENT BIOCOMPATIBILITY MICRO-PARTICULATE SHEDDING.

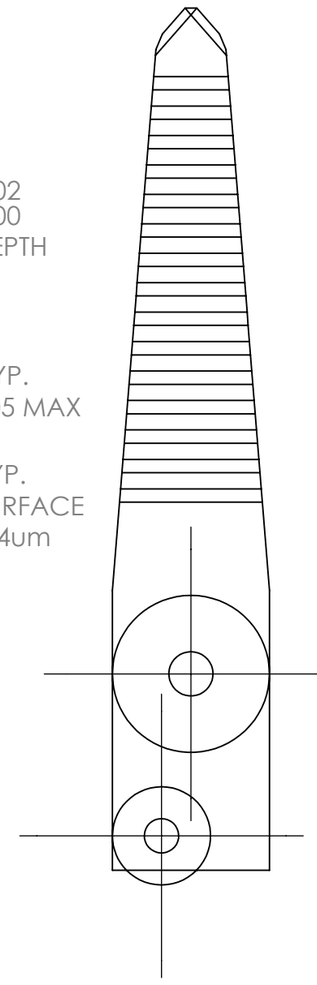
3. MICRO-MACHINING & GD&T CONTROLS:
3.1 PRIMARY DATUM [A] (SHEAR MATING STEP FACE): FLATNESS <= 0.003 mm, SURFACE FINISH Ra <= 0.2 MICRONS.
3.2 PIVOT BEARING BORE [B]: DIA 0.900 +0.008/-0.000 mm (H7), PERPENDICULARITY <= 0.005 mm RELATIVE TO DATUM [A].
3.3 ACTUATION LINKAGE BORE: DIA 0.700 +0.010/-0.000 mm, TRUE POSITION WITHIN 0.015 mm RELATIVE TO [A][B][C].
3.4 GRASPING TEETH: ATRAUMATIC TOOTH PROFILES (PITCH 0.60 mm, DEPTH 0.15 mm). UNIFORMITY ACROSS JAW LENGTH.
3.5 NECK TRANSITION: BLEND RADIUS R 0.30 +/- 0.05 mm CONTINUOUS AND TANGENT WITHOUT TOOL CHATTER OR DWELL MARKS.

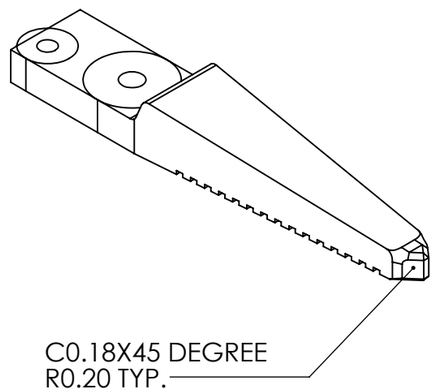
4. DEBURRING & ATRAUMATIC EDGE REQUIREMENTS:
4.1 CRITICAL DEBURRING: 100% BURR-FREE INSPECTION UNDER 30X OPTICAL MICROSCOPY. NO LOOSE PARTICLES, FLASH, OR FEATHER EDGES PERMITTED.
4.2 ATRAUMATIC RADIUS: ALL EXPOSED OUTER JAW EDGES AND FRONT TIP MUST HAVE CONTINUOUS ATRAUMATIC BLENDING (R 0.10 - 0.20 mm) TO PREVENT TISSUE LACERATION DURING MINIMALLY INVASIVE SURGICAL (MIS) PROCEDURES.

5. SURFACE FINISHING & CLEANROOM CLEANLINESS:
5.1 SURFACE FINISH: ELECTRO-POLISHED (EP) IN ACCORDANCE WITH ASTM F86 TO PASSIVATE SURFACES AND ENHANCE BIO-STABILITY (Ra <= 0.2 MICRONS ON BEARING & SHEAR FACES, Ra <= 0.4 MICRONS ON EXTERIOR).
5.2 CLEANLINESS PROTOCOL: MULTI-STAGE ULTRASONIC CAVITATION CLEANING IN PURIFIED DEIONIZED WATER FOLLOWED BY CLEANROOM PACKAGING COMPLIANT WITH ISO 14644-1 CLASS 4.









UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: Linear ±0.02 mm, Angular ±0.2°	FINISH: Electropolished (EP) per ASTM F86, Ultrasonic Cleaned				DEBURR AND BREAK SHARP EDGES	DO NOT SCALE DRAWING / REVISION: A
						ASME Y14.5-2018 / ISO 8015 / ISO 2768-m (Fine for Micro-Features)
	NAME	SIGNATURE	DATE			TITLE: SURGICAL ROBOT ATRAUMATIC MICRO-GRASPER JAW (3mm CLASS MIS)
DRAWN						
CHK'D						
APPV'D						
MFG						
Q.A				MATERIAL: Ti-6AL-4V ELI (ASTM F136 / ISO 5832-3) OR 316LVM (ASTM F138 / ISO 5832-1)		DWG NO. W3-SURG-GRASPER-5.1 / A3
				WEIGHT: ~0.18 g (Ti-6AL-4V) / ~0.32 g (316LVM)		SCALE: NTS / SHEET 1 OF 1